2110211 Introduction to Data Structure

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Lecture 13

Lecture 1

C++ Intro

variable

C++ from Java & Python

data

C++, if, for, while, function

scale

qwertyuio

Lecture 2

Array & Vector

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
  vector<int> a = {2,3};
  vector<bool> b = {true,false,true};

  a.push_back(10);
  a.push_back(20);

  a.insert( a.begin(), 99);
  a.insert( a.end(), -5);

  sort(a.begin(),a.end());

  cout << "a is ";
  for (size_t i = 0; i < a.size(); i++) {
    cout << a[i] << " ";
  }
  cout << endl;

  cout << "b is ";
  for (size_t i = 0; i < b.size(); i++) {
    cout << b[i] << " ";
  }
  cout << endl;
}
```

Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
```

```
#include <vector>
#include <set>
#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
  string filename = "test.txt";
  printWord(filename);
  printUniqueWords1(filename);
  printUniqueWords2(filename);
  printUniqueWords3(filename);
  printUniqueWords4(filename);
  return 0;
}

bool search(string words[], int n, string w) {
  for (int i = 0; i < n; i++) {
    if (words[i] == w) return true;
    n++;
  }
}

void printWord(string filename) {
  int n = 0;

  Tokenizer tokenizer(filename);
  while(tokenizer.hasNext()) {
    string token = tokenizer.next();
    n++;
  }
  tokenizer.close();
  cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
  string words[100000];
  int n = 0;

  Tokenizer tokenizer(filename);
  while(tokenizer.hasNext()) {
    string token = tokenizer.next();
    if (!search(words,n,token)) words[n++] = token;
  }
  tokenizer.close();
  cout << "A total of " << n << " words" << endl;
}
```

Dynamic Array

```
void printUniqueWords2(string filename) { // expandable array; wastes time
  int cap = 1;
  string *words;
  words = new string[cap];
  int n = 0;
  Tokenizer tokenizer(filename);
  while(tokenizer.hasNext()) {
    string token = tokenizer.next();
    if (!search(words,n,token)) {
      if (n == cap) {
        string *a = new string[2*cap];
        for (int i=0; i<n; i++) a[i] = words[i];
        delete[] words;
        words = a;
        cap *= 2;
      }
      words[n++] = token;
    }
  }
  tokenizer.close();
  cout << "A total of " << n << " words" << endl;
}
```

Word Count by Vector & Set

Vector

```
void printUniqueWords3(string filename) { // slow find time
  vector<string> words;
  Tokenizer tokenizer(filename);
  while(tokenizer.hasNext()) {
    string token = tokenizer.next();
    if (words.end() = find(words.begin(), words.end(), token))
      words.push_back(token);
  }
  tokenizer.close();
  cout << "A total of " << n << " words" << endl;
}
```

Set

```
void printUniqueWords3(string filename) { // quick find existence
  set<string> words;
  Tokenizer tokenizer(filename);
  while(tokenizer.hasNext()) {
    string token = tokenizer.next();
    // if (words.end() == find(words.begin(), words.end(), token)) // slow find
    if (words.end() == word.find(token)) // if can't find then return word.end()
      words.insert(token);
  }
  tokenizer.close();
  cout << "A total of " << n << " words" << endl;
}
```

Lecture 3

std::pair

Basic

```
#include <iostream>

int main() {
  std::pair<int,float> x;
  std::pair<int,float> y;

  x.first = 10;
  x.second = 20.65;

  std::cout << x.first << " "; // 10
  std::cout << x.second << std::endl; // 20.65

  y = x;

  std::cout << y.first << std::endl; // 10
  std::cout << y.second + 20 << std::cout; // 40.65
}
```

Pair reserves 2 spaces in RAM for `first` and `second` elements. When you use `y = x`, it copies values of `x` to `y`. You need to build template by specifying data types in a pair.

Initialize

```
#include <iostream>

int main() {
  // default constructor
  std::pair<std::string,bool> p; // default of that type: {"", false}
  std::cout << "default [" << p.first "] [" << p.second << "]" << std::endl;

  // initialize by { }
  std::pair<std::string,bool> p1 = { "somchai", true };

  // create pair without specifying type by "make_pair"
  std::pair<bool,int> p2;
  p2 = std::make_pair(false,10);

  std::pair<bool,int> p3(p2); // must be same type p3 = p2

  // more complex pair
  std::pair< std::pair<float,int>, std::string >  p4 = { {20.5, -3}, "abc"};
  std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}
```

Array inside Pair

```
#include <iostream>

int main() {
    std::pair<std::string*,std::pair<int, float>> p;
```

```
    p.first = new std::string[5];
    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10,1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}
```

Template

- Template allows "same code, different data types"
- `std::pair` is a "class template"
  - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply "Type information" to the template what `T1` and `T2` should be.

STD and Namespace

- The "Fullname" of cout is `std::cout`
- `std` is a namespace
- We need to use fullname
  - Too lazy? use "`using`" keywords

```
#include <iostream>

// this tells C++ that when we say cout, we mean std::cout;
using std::cout;

int main() {
  // we still need to use std::endl
  cout << "Yes" << std::endl;
}
```

```
#include <iostream>

// this tells C++ that when we say something that is does not understand, C++ should
try to use std as its namespace

// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;

int main() {
  cout << "Yes" << endl;
}
```

std::vector

A linear storage of a single data type

Basic

```
#include <iostream>
#include <vector>

using namespace std;
```

```
int main() {
  vector<int> v1;
  cout << "Size of v1 is " << v1.size() << endl; // 0

  vector<int> v2 = {2,3,4};

  cout << v2[1] << endl; // 3
  v1 = v2;
  v1[0] = 20;

  cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

  v1.push_back(99);
  cout << v1.size() << endl;
}
```

- Vector starts with empty element
- Use `size()` to get the number of element
- Use `empty()` to check if a vector has any element

Initialize

- With specific size
- With specific size and starting value

```
#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
  for (size_t i = 0; i < v.size(); i++) {
    cout << v[i] << " ";
  }
  cout << endl;
}

int main() {
  vector<float> v1(10); // size = 10
  print_vector(v1);

  vector<float> v2(5,3.55); // size = 5, default = 3.55
  print_vector(v2);

  vector<float> v3(v2);
  print_vector(v3);
}
```

Vector Access & Overflow Problem

Access

```
#include <iostream>
#include <vector>

using namespace std;

int main() {
```

```
  vector<float> v1(2);
  vector<float> v2(2);

  cout << "-- v1 --" << endl;
  for (int i = 0; i < 7; i++) {
    v1[i] = i; // this automatically assigns value
    cout << i << ": " << v1[i] << endl; // print nonsense if out of range
  }
  cout << "-- v2 --" << endl;
  for (int i = 0; i < 7; i++) {
    cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
space of v1)
  }
  cout << "using at" << endl;
  v2.at(1) = 99;
  // this will cause exception 'std::out_of_range'
  for (int i = 0; i < 7; i++) {
    cout << ": " << v2.at(i) << endl;
  }
}
```

- Operator `[]` won't check for 'out-of-range'
  - Reading, writing beyond size is undefined
  - Might crash
  - Grader will give 'x'
- at `()` will check bound
  - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

Modifying Vector

Resizing

- Resize change the size
  - Enlarge will fill with default

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
  cout << "Size of V is " << v.size() << ":";
  for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
  cout << endl;
}

int main() {
  vector<int> v(3,10);
  print(v); // 10,10,10,
  v.resize(6);
  print(v); // 10,10,10,0,0,0,
  v.resize(1);
  print(v); // 10,
  v.resize(7);
  print(v); // 10,0,0,0,0,0,0
}
```

Modify

- pop_back
  - Erase last element
- insert(position,value)
- erase(position)
- Careful!
  - Both insert and erase position must be valid
  - If not vald, it can crash

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
  cout << "Size of V is " << v.size() << ": ";
  for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
  cout << endl;
}

int main() {
  vector<int> v(3,8); // v.begin() is called iterator
  v.insert( v.begin(), 1);    // 1, 8, 8, 8
  v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8
  v.insert( v.end(), 3);     // 1, 8, 8, 2, 8, 3
  print(v);
  v.erase(v.begin();          // 8, 8, 2, 8, 3
  v.erase(v.begin()+2);       // 8, 8, 8, 3
  print(v);
  v.pop_back();               // 8, 8, 8
  print(v);
}
```

std::find & std::sort

Find

- find(a,b,c)
  - a and b are position (iterator)
  - find c from a to *before* b
  - if not found return b
- Must #include <algorithm>

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
  vector<int> v = {9,-1,3,7,5,2,1,4};

  int x = 5;
  if ( find(v.begin(), v.end(), x) != v.end() ) {
    cout << "found" << endl;
  } else {
    cout << "not found" << endl;
```

```
  }

  if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;

  // How many "YES" will be printed? (CHEAT QUESTION)
  if (find(v.begin()  , v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
  if (find(v.begin()  , v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES
  if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
  if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES
  // First 3 can't even give v.end(), last did find so it gave that position instead
}
```

Sort

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

void print(vector<int> v) {
  cout << "Size of V is " << v.size() << ": ";
  for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
  cout << endl;
}

int main() {
  vector<int> v = {9,-1,3,7,5,2,1,4};

  print(v);  // 9, -1, 3, 7, 5, 2, 1, 4,
  sort(v.begin()+2, v.begin()+6);
  print(v);  // 9, -1, 2, 3, 5, 7, 1, 4,
}
```

Functions that Work with Vector

- `sort`
- `find`
- `min_element`
- `max_element`
- `lower_bound`
- `upper_bound`

Need to use *iterator*

Lecture 4

Vector Iterator

Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`
- For `insert`, valid position is from `begin()` to `end()`, inclusive
- For `erase`, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator

  - v1.end() is not the same as v2.end()

Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (++) and decrement (--)

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
  vector<int>   v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
  vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

  vector<int>::iterator it1 = v1.begin() + 3; // 30
  vector<float>::iterator it2 = v2.end();      // after 2.73

  // getting value at iterator by using "*" operator
  cout << *it1 << endl;      // 30
  cout << *(it2-1) << endl; // 2.73
  cout << *it1+2 << endl;    // 32

  // iterator arithmetics
  vector<int>::iterator it3 = it1 + 2;
  cout << "data at it3: " << *it3 << endl; // 50
  cout << "difference of it3,it1: " << (it3 - it1) << endl; // 2

  vector<float>::iterator it4 = v2.end();
  it4--;
  cout << "data of it4: " << *it4 << endl; // 2.73

  // this cannot be done
  // cout << (it2 - it1) << endl;
}
```

Iterator All Elements by Iterator

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
  vector<int>   v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
  vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

  cout << "----v1----" << endl;
  auto it = v1.begin();
  while (it < v1.end()) { // should use != in other iterator types
    cout << it - v1.begin() << ": " << *it << endl;
    it++;
  }

  cout << "---v2---" << endl;
  for (auto it = v2.begin(); it < v2.end(); it++) {
    cout << it - v2.begin() << ": " << *it << endl;
```

```
  }
}
```

- We can compare iterator
  - Beware!! Iterator of some data structure does not support > or < comparator
  - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

Iterator Invalidate

Some Functions that Use Iterators

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
  vector<int>   v1 = {3,0,1,2,4,-3,9,8};
  vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

  auto it1 = min_element(v1.begin(),v1.end());
  auto it2 = max_element(v2.begin()+2,v2.end());

  cout << *it1 << endl; // -3
  cout << *it2 << endl; // 3.14

}
```

- `min_element` and `max_element` get the iterator of min, max element

New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
  vector<string> v1 = {"somchai", "somying", "somsak"};

  // range-based for loop
  for (string x : v1) {
    // x is a copy of each element in v1
    cout << x << ", ";
  }

  // using reference
  // x is THE element of v1, meaning we can modify it
  for (auto &x : v1) { x.replace(0,4,"--"); }
  for (auto &x : v1) { cout << x << " ";} // --hai --ing --ak
  cout << endl;
}
```

Iterator Invalidation

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
  vector<int> v1 = {10,20};

  auto = v1.end()-1; // this points to 20
  // we resize (enlarge) v1, now [it] is invalidated
  v1.resize(10);

  // this might not crash but it actually points to somewhere not in v1
  cout << *it << endl;

  // this will crash the program
  // v1.insert(it,99);
  for (auto &x: v1) {cout << x << " ";}
}
```

- Some operation on the data structure *invalidate* existing iterators
  - For vector, this includes *addition* and *deletion* of element (including *resize*)

Set

- Storing distinct data of same type
  - The data type must be *comparable*, i.e., we can tell if a is more or less than be
- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from "minimum element" and goes in increasing value direction
  - Can be used to (somewhat) fast storing

Basic

- Notice that `s` does not include duplicate elements
- Also see that when we iterate, member is sorted
  - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
  set<int> s = {4,1,3,2,1,1,3,4};

  cout << "Size of s is " << s.size() << endl; // 4

  s.insert(10);
  s.insert(5);
  s.erase(3);

  cout << "member of s: "; // 1 2 4 5 10
  for (auto it = s.begin(); it != s.end(); it++) {
    cout << *it << " ";
  }
}
```

Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
  - For now, please believe that
    - If there are N elements in the set
    - Insert takes time directly proportional to $\log(N)$
    - Each `it++` or `it--` takes times directly proportional to $\log(N)$
    - Each find takes time directly proportional to $\log(N)$

Set Iterator

Demos Comparing Vector & Set

- Insert
  - Vector is faster that set
- Find random number
  - Set is much faster than vector

Set Iterator

- We cannot do `s.begin() + x`
  - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
    - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < 0; i++) {it++;}`
  - We cannot *compare by > or <*
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or x

```
#include <iostream>
#include <set>
using namespace std;

int main() {
  set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
    {"z",0}, {"z",-1}, {"z",9} };

  for (auto &x : s) {
    cout << x.first << "," << x.second << endl;
  }

  cout << "-- find --" << endl;
  auto it = s.find({"z",-1});
  cout << (*it).first << "," << (*it).second << endl;
  it--;
  it--;
  cout << it->first << "," << it->second << endl;
  it++;
  cout << it->first << "," << it->second << endl;
}
```

Additional Function

- `set.lower_bound`
- `set.upper_bound`
- `set.count`

std::map

Associative data structure with same property as set

Map

- Is very similar to Python's `dict` in usage
- Is internally implemented as a set with "pair" data type
  - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

Basic

```
#include <iostream>
#include <map>
using namespace std;

int main() {
  // map between "Key Type" string and "Mapped Type" int
  map<string,int> m;
  m["somchai"] = 10;
  m["somying"] = -5;
  cout << "Size = " << m.size() << endl; // 2

  // accessing unseen key create a map with default value
  cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
  // each element is a pair of key type and mapped type
  for (auto it = m.begin(); it != m.end(); it++) {
    cout << it->first << " is mapped to " << it->second << endl;
  } // somchai: 10, somying: -5

  // this will create mapping "abc" to 0 first and then increase it
  m["abc"]++;
  cout << "now size = " << m.size() << endl; // 4
  for (auto &x : m) {
    cout << x.first << " is mapped to " << x.second << endl;
  } // abc: 1, somchai: 10, somying: -5, xxx: 0
}
```

Checking If Map Has This Key?

- Use `find()`

```
#include <iostream>
#include <map>
using namespace std;

int main() {
  map<int,string> m;
  m[1] = "somchai";
  m[99] = "nattee";

  int k = 99;
  map<int,string>::iterator it;
  if ((it = m.find()) != m.end()) {
    cout << "Key " << it->first << " is mapped to " << it->second << endl;
  } else {
    cout << "Key " << k << " does not exist in m." << endl;
  }

  // this is not the correct way to check if key exists why??
  if (m[k] != "") {
```

```
    cout << "exists" << endl;
  } else {
    cout << "does not exists" << endl;
  }
}
```

Requirement of std::set and std::map

- Set data type and map key type must be comparable
  - We must be able to compare *order* of two elements
- Type that we can use directly
  - int, bool, float, string, double, char, … and most of other numerical data type
  - Pair can also be used if both the types of first and second are comparable
    - Pair compare first then second

Practice Reading C++ Docs

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
  - For `set<int> s`, can we do `s.erase(20)`
  - For `map<string,bool> m`, can we do `m.erase("Somchai")`??
- If we wish to erase element from index 3 to index 4096 in a vector
  - Is there any function from vector that we can easily use?

Pair-Sum Problem

- Pair Sum
  - Given an array of integers, out task is to find whether there exists a pair of elements in the array such that their summation equal to X
- Input:
  - Array of integers (our main array); this is a large array
  - M values of X, for each value X, we have to determine if a pair whose sum equal to X exists.
- Output:
  - For each value of X, print “YES” if we found such pair; print “NO” otherwise

Answer

```
#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
  int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
  int n = sizeof(a) / sizeof(a[0]);
  int X = 10000;

  cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
  for (int i=0; i<(pow(2,n)); i++) {
    int t = X;
    for (int j=0; j<n; j++) {
```

```
      if (i / ((int) pow(2,j+1)) % 2 == 0) {
        t -= a[j];
      }

      if (0 == t) {
        cout << "YES" << endl;
        return true;
      }
      if (0 > t) {break;}
    }
  }
  cout << "NO" << endl;
  return false;
} // My stupid solution where I brute force every possible combinations
```

Lecture 5

Stack

การเพิ่ม / ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last_In First-Out)

กองซ้อน : stack

```
bool            empty();
unsigned        size();
T               top();            // ดูบนสุด
void            push(T element);  // ใส่ข้อมูล
void            pop();            // เอาออก
```

ข้อควรระวัง

- ใช้ `#include <stack>`
- `top()` ตอนไม่มีข้อมูล → พัง
- `pop()` ตอนไม่มีข้อมูล → พัง
- stack ไม่มี iterator
  - ดูได้เฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (`top`)
  - ไม่มี `begin()`, `end()` ให้ใช้
  - ถ้าอยากมาดูข้อมูลทุกตัวใน stack ต้องจำใจ pop ข้อมูลออกมา

ตัวอย่างการใช้งาน Stack

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ `(){}[]`...
- การจัดเก็บตัวแปรและการทำ function calls
- การประมวลผลนิพจน์ทางคณิตศาสตร์ postFix → Discrete Math
- การทำ undo/redo
- การค้นคำตอบแบบ depth-first search
- ...

Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : `( { ( ) [ { } ] } )`
- ผิด : เปิดปิดไม่ตรงกัน `( { ] )`
- ผิด : มีปิดมากเกินไป `( { ( ) } ) ) }`

- ผิด : มีเปิดมากเกินไป `( { ( ) }`
- วิธีทำ
  - อ่านมาทีละตัว
  - ถ้าเป็นวงเล็บเปิด ให้ push ลง stack
  - ถ้าเป็นวงเล็บปิด ให้ pop จาก stack มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
  - เมื่อใดที่อยาก pop ถ้า `isEmpty` แสดงว่า ปิดมีมากไป
  - เมื่ออ่านเสร็จหมด stack ยังมีข้อมูล แสดงว่า เปิดมีมากไป

Function Call Stack

Using Stack in Java Virtual Machine

- jvm uses java stack to store
  - state of method calls
  - parameters and local variables
  - temporary memory for computation (operand stack)

```java
public class StackFrame {
  public static void main(String[] args) {
    a(3,2);
    b(5);
  }
  static void a(int x, int y) {
    int z = x/y;
    b(z);
  }
  static void b(int x) {
    ++x;
  }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function `main` contains `args` and RA `JVM` and calls `a`, `a` will be added to stack and contains `z, y, x`, and RA `04:`, then it calls `b`. After `b` is finished, it will come back to line 9 and popped itself. It will do `a` and go to line 4, and so on. Arguments in each functions are separated variables.

```cpp
#include <iostream>

using namespace std;

void test(int x) {
  if (x > 0) {
    int y = x-1;
    cout << "I am test(" << x << "). continue... " << endl;
    test(y);
    cout << "now x is " << x << endl;
  } else {
    cout << "I am test(0). stop"
  }
}

int main() {
  test(4);
  return 0;
}
```

Output

```
I am test(4). continue...
I am test(3). continue...
I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
now x is 3
now x is 4
```

Postfix Evaluation

Infix and Postfix Expressions

- infix (เติมกลาง)
  - `a + b * c / d - 2`, `(a + b) * c / (d - 2)`
  - use order of operations
  - use parentheses
- postfix (เติมหลัง)
  - `a b c * d / + 2 -`, `a b + c * d 2 - /`
  - order is from left to right
  - parentheses may be omitted

```
1  2  +  3  *  4  1  -  /
      3  3  *  4  1  -  /
            9  4  1  -  /
            9        3  /
                        3
```

Calculate Postfix Expressions with Stack

- Value of `2 3 + 4 5 - 6 * +` ?
- Use stack to help
- Solution
  - Check each element in postfix from left to right
  - If it is operand, push stack
  - If it is operator, pop elements from stack as many as operator needs, then push the result
  - After complete, answer will be at the top of stack

Convert Infix to Postfix

Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
  - Read each element in infix
  - If it is operand, move it to the back of output
  - If it is operator
    - May pop operator to the end of output
    - push operator to stack
  - If all infix is read
    - pop all operators to the end of input

```
string infix2postfix(string &infix) {
  int n = infix.length();
  string postfix = "";
  stack<char> s;
```

```
  for (int i=0; i<n; i++) {  // Read each element
    char token = infix[i];
    if (token เป็น operand) {
      postfix += token;  // if operand, add to output
    } else {
      while ( ???? ) {
        postfix += s.top();  // if operator, pop stack to result,
        s.pop();
      }
      s.push(token);         // then push new operator
    }
  }
  while (!s.empty()) {postfix += s.top(); s,pop();}
  return postfix;
}
```

Priority of Operators

```
input   2   *   3   +   4

read      output     stack
 2
 *          2
 3          2          *
 +         2 3         *   (* comes before and is more important than +)
 4        2 3 *        +
         2 3 * 4       +
        2 3 * 4 +
```

```
input   2   +   3   *   4

read      output     stack
 2
 +          2
 3          2          +
 *         2 3         +
           2 3        * +
          2 3 *        +
         2 3 * +
```

Code to Compare Priority

```
string infix2postfix(string &infix) {
  int n = infix.length();
  string postfix = "";
  stack<char> s;
  for (int i=0; i<n; i++) {
    char token = infix[i];
    if (priority.find(token) != priority.end()) {
      postfix += token;
    } else {
      int p = priority[token];  // Get priority of the new operator
      while ( !s.empty() && priority[s.top()] >= p ) {  // lazy evaluation; left first
        postfix += s.top();  // pop if top operator is not less important than new op
        s.pop();
      }
      s.push(token);
    }
```

```
  }
  while (!s.empty()) {postfix += s.top(); s,pop();}
  return postfix;
}
```

Lazy Evaluation

Setting Priority of Operators

- ^ represents power 7
- ^ before + - * /
- * / before + -
- * same as / 5
- + same as - 3

```
map<char,int> priority =
    {{'+',3},{'-',3},{'*',5},{'/',5},{'^',7}};
```

Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

```
input   2   *   (   3   +   4   )

read      output     stack
 2
 *           2
 (           2          *
 3           2         ( *
 +          2 3        ( *
 4          2 3      + ( *
 )         2 3 4     + ( *   (pop all)
        2 3 4 + *
```

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
  postfix += s.top();
  s.pop();
}
s.push(token);

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
  postfix += s.top();
  s.pop();
}
if (token == ')') s.pop(); else s.push(token);
```

Priority of Operators with Parentheses

	+	-	*	/	^	(	)
In expression	3	3	5	5	7	9	1
In stack	3	3	5	5	7	0	

Right to Left Operator

	+	−	*	/	^	(	)
In expression	2	2	4	4	8	9	1
In stack	3	3	5	5	7	0	

Note Power in expression is more important than in stack

```
input   2   +   3   -   4   ^   5   ^   6

read     output     stack
 2
 +          2
 3          2          +
 -         2 3         +
 4        2 3 +        -
 ^       2 3 + 4      ^ -
 5       2 3 + 4      ^ -
 ^      2 3 + 4 5     ^ -
 6      2 3 + 4 5    ^ ^ -
       2 3 + 4 5 6   ^ ^ -
    2 3 + 4 5 6 ^ ^ -
```

Conclusion

- Implementations of stack to solve problems
- Main operations: `push` / `pop` / `top`
- Make stack with array
- If reserve enough space, all runs is $\Theta(1)$; constant time

Lecture 6

Queue

- Just like a normal queue in real life.
  - First-in-First-Out data structure
  - One way in (`back` of queue), one way out (`front` of queue)
  - `push` = add data to the back of the queue
  - `pop` = remove data from the head of the queue

Basic

```cpp
#include <iostream>
#include <queue>
#include <vector>

using namespace std;

int main() {
  queue<int> q;
  q.push(1);
  q.push(2);
  q.push(3);
  while (q.empty() == false) {
    cout << q.front() << endl;
```

```
    q.pop();
  }
  cout << "-- example 2 --" << endl;
  queue<vector<int>> q2;
  vector<int> v1 = {1,2,3};
  vector<int> v2 = {99,88,-1};
  q2.push( v1 );
  q2.push( v2 );
  cout << q2.back()[1] << endl;
  cout << q2.front().size() << endl;
  auto x = q2.front();
  q2.pop();
  cout << x[0] << endl;
}
```

```
size_t      q.size()
bool        q.empty()
void        q.push(T data)
void        q.pop()
T           q.front()
T           q.back()
```

Limitation

- Same limitation as stack
  - No iterator
    - No `begin()`, `end()`
    - Can only access front and back of the queue
    - If we wish to access all members, we have to pop it all
  - Do not call `front()`, `back()`, `pop()` when the queue is empty

Radix Sort

Queue Application: Fast sorting with no comparison

Overview

- Put all data in an array
- For each digit X, from LSD to MSD
  - PUT TO QUEUE step: Sort by digit X by putting all of data from the array into B queues
    - B is the base of the number
    - For example, for a base 10 number, we will have `Queue[0]` to `Queue[9]`
    - Put into the queue labelled with that digit
  - GET FROM QUEUE step: Start form queue 0 to queue $B - 1$, remove data from the queue and put back to the array

Example

Input 115 15 42 305 21 8 463

Round 1 (digit 0), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
					305				
					15				
	21	42	463		115			8	

$\Rightarrow$ 21 42 463 115 15 305 8

Data is now sorted by the last digits, bacause we pop from queue 0 to 9

Round 2 (digit 1), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
8	15								
305	115	21		42		463			

⇒ `305 8 115 15 21 42 463`

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

Round 3 (digit 2), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
42									
21									
15									
8	115		305	463					

⇒ `8 15 21 42 115 305 463`

Code

```
#define base 10

int getDigit(int v, int k) {
  // return the kth digit of v (MSD is digit 0)
  int i;
  for (int i=0; i<k; i++) v /= base;
  return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
  queue<int> q[base];
  for (int k=0; k<dl k++) {
    for (auto &x : data) {
      q[getDigit(x,k)].push(x);
    for (int i=0,j=0; i<base; i++) {
      while(!q[i].empty()) {
        data[j++] = q[i].front(); q[i].pop();
      }
    }
    }
  }
}
```

Breadth First Search

Queue Application: Gotta generate 'em all

The Problem

- Given a positive integer X
- Start with a number 1, find a sequence of arithmetics operations, either "`* 3`", or "`/ 2`" that makes 1 into X
  - the `/ 2` is integer division, e.g., `5 / 3 = 1` (not 1.6666)
  - The sequence must be as short as possible
- Example
  - `10 = 1 * 3 * 3 * 3 * 3 / 2 / 2 / 2`
  - `31 = 1 * 3 * 3 * 3 * 3 * 3 / 2 / 2 / 2 / 2 / 2 * 3 * 3 / 2`

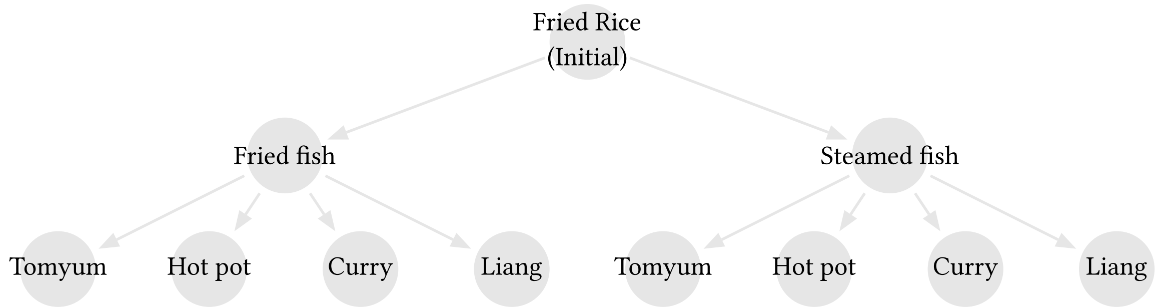
The Idea

- Generate all possible sequences
  - Start with length $1, 2, 3, \ldots$ until we find one that gives X
- This is called an *exhaustsive search* algorithm
  - Systematically enumerate all possible somethings

Tree Structure (Search Tree)

- A structure to illustrate *search* algorithm
- Divide into steps
  - Start with *intial solution*
  - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
    - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

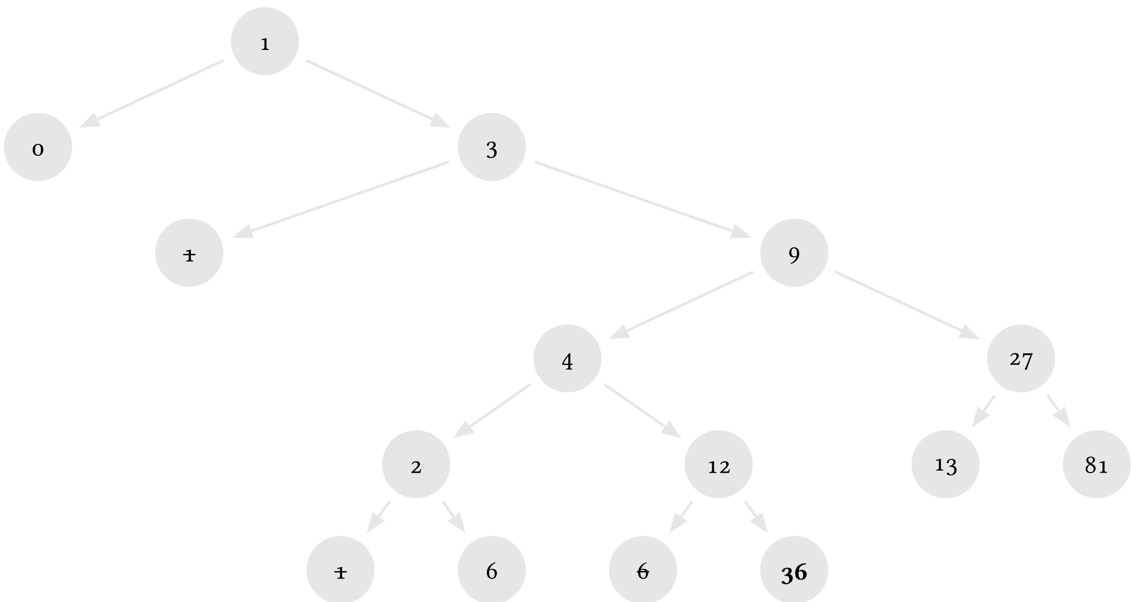
Enumerate



M3D2 Problem & Solution

Back to Our Problem

- Start with 1
- Each step is either `* 3` or `/ 2`
- Issue: might get repeated number
  - Solution: if we have found it, do not generate new step



M3D2 Code

Code

- Queue makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```
void m3d2(int target) {
  map<int, int> prev;
  queue<int> q;
  int v = 1;
  q.push(1); prev[1] = -1;
  while (!q.empty()) {
    v = q.front(); q.pop();
    if (v == target) break;
    int v2 = v/2;
    int v3 = v*3;
    if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
    if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
  }
  if (v == target) showSolution(v, prev);
}
```

Display Solution

x	0	1	2	3	4	6	9	12	13	27	36	39	40
prev[x]	1	-1	4	1	9	2	3	4	27	9	12	13	81

showSolution

```
void showSolution(int v, map<int,int>& prev) {
  string out = "";
  while (prev[v] != -1) {
    if (prev[v] * 3 == v) {
      out = "x3" + out;
    } else {
      out = "/2" + out;
    }
    v = prev[v];
  }
  out = "1" + out;
  cout << out << endl;
}
```

⇒ `36 = 1 * 3 * 3 / 2 * 3 * 3`

Lecture 7

Priority Queue

Queue with privilege

Intro

- Priority Queue is …
  - A queue with priority
  - Item with high priority is promoted to the front of the queue
    - There is *no back of the queue*

    - Priority is defined by having more value
        - Comparison, by default, is to use `operator <`, i.e., if item `A < B` is `true`, then `B` *has higher priority*
        - We can have custom comparator
- Has the *same interface* as stack

Example

For intuitive purpose only! This is not really how priority_queue work internally

```
push(10)
  < 10         >

push(30)
  < 30 10      >

push(20)
  < 30 20 10   >

pop()
  < 20 10      >

push(15)
  < 20 15 10   >
  |
 top
```

Basic

```
size_t       q.size()
bool         q.empty
void         q.push()
void         q.pop()
T            q.top()
* there is no q.back()
```

Limitation

- Same limitation as stack, queue
    - *no iterator*
        - No `begin(), end()`
        - Use `#include <queue>`
        - Can only access `top()` of the queue
        - If we wish to access all members, we have to pop it all
    - Do not call `top(), pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

Class

Quick Summary

- Syntax
    - class declaration must end with `;`
    - Function definition can be outside of the class
    - Access modifier is `public:, private:, protected:`
    - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
    - When declared, a constructor is called

Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
  void setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
  }
  string getFullName() {
    return "[" + name + " " + surname + "]";
  }
private:
  string name, surname;
};

int main() {
  Student a;
  Student b;
  a.setFullName("nattee", "niparnan");
  cout << a.getFullName() << endl;
  cout << b.getFullName() << endl;
}
```

Declaring class before writing functions

```
class Student {
public:
  void setFullName(string name, string surname);
  string getFullName();
private:
  string name,surname;
};

void Student::setFullName(string name, string surname) {
  this->name = name;
  this->surname = surname;
}

string Student::getFullName() {
  return "[" + name + " " + surname + "]";
}
```

Output

```
[nattee niparnan]
[ ]
```

Example 2: Constructor

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
```

```
  Student(float score) {gpax = score;}
  void setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
  }
  string getFullName() {return "[" + name + " " + surname + "]";}
  bool is1stHonor() {return gpax >= 3.6;}
private:
  string name, surname;
  float gpax;
};

int main() {
  Student a(2.95);
  a.setFullName("nattee", "niparnan");

  cout << a.getFullName() << endl;
  if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
  // Student b; // <-- cannot compile because there is no default constructor
}
```

Operator Overloading

How C++ has a function for each operator

Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
  - This can be considered the same as calling a function `plus(a,b)`
  - C++ allows us to write a function for many operator and use it as an operator
  - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```
Matrix multiply(Matrix a1, Matrix a2) {
  multiply(a, multiply(b,c))
}
```

We want to write

```
Matrix a, b, c
a * (b * c)
```

Example

```
#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
  string result = "";
  for (int i=0; i<rhs; i++) {
    result = result + lhs;
  }
  return result;
}
```

```
int main() {
  string a = "abc ";
  cout << (a * 3) << endl; // this gives "abc abc abc "
}
```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+, -, *, /, %`)
- Some takes one (such as `++, --, !, *, &`)

Overloading less-than operator

Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
  - Let us consider operator overloading

Overloading <

- As stated earlier, set, map, and priority_queue use *operator* < to compare two elements
- It does not work if we overload *operator* >

```
class Student {
public:
  Student(float score, string a, string b) {
    name = a;
    surname = b;
    gpax = score;
  }
  bool is1stHonor() {return gpax >= 3.6;}
  // not good, now our data is public
  string name, surname;
  float gpax;
  // overloading <
  bool operator<(const Student & other) const {
    return gpax < other.gpax;
  }
};

int main() {
  Student a(2.95,"nattee","niparnan");
  Student b(4.00,"attawith","sudsang");
  cout << (a < b) << endl; // 1
  priority_queue<Student> pq;
  pq.push(a);
  pq.push(b);
  cout << pq.top().name << endl; // attawith
}
```

Custom Comparator

Why custom?

- By overloading *operator* <, we have defined default ordering of that class
- What if we need another ordering, just for this *priority_queue* only

- For example, *Student* is ordered by *gpax* by default
- What if we want our priority_queue to order by name instead, while keeping the *Student* default ordering elsewhere
- Better, can we have multiple priority_queue with different ordering?

- Can be done via comparator class

Example

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
  bool operator()(const Student& lhs,
                  const Student& rhs) {
    return lhs.name < rhs.name;
  }
};

class GpaxThenName {
public:
  bool operator()(const Student& lhs,
                  const Student& rhs) {
    if (lhs.gpax == rhs.gpax)
      return lhs.name < rhs.name;
    return lhs.gpax < rhs.gpax;
  }
};

int main() {
  Student a(2.95,"nattee","niparnan");
  Student b(4.00,"attawith","sudsang");
  Student c(4.00,"vishnu","kotrajaras");
  cout << (a < b) << endl;  // 1
  StudentByNameComparator comp1;
  GpaxThenName comp2;

  cout << comp1(a,b) << endl; // 0; can use like function

  // Use 3 template parameters
  priority_queue<Student,          // class to compare
                 vector<Student>,  // vector of that class
                 StudentByNameComparator> pq(comp1); // comparator class
  pq.push(a);
  pq.push(b);
  cout << pq.top().name << endl;  // nattee

  priority_queue<Student,
                 vector<Student>,
                 GpaxThenName> pq2(comp2);
  pq2.push(a);
```

```
  pq2.push(b);
  pq2.push(c);
  cout << pq2.top().name << endl;  // vishnu
}
```

Another Method, lambda-function

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
  auto compare = [](const string& lhs, const string& rhs) {
    return lhs.size() < rhs.size();
  };

  cout << "Result of compare function = " << compare("xxx","z") << endl;

  priority_queue<string,vector<string>,decltype(compare)> pq(compare);
  pq.push("somchai");
  pq.push("z");
  pq.push("abc");
  while (pq.empty() == false) {
    cout << pq.top() << endl;
    pq.pop();
  }
}
```

Output

```
Result of compare function = 0
somchai
abc
z
```

Priority Queue Template

Templating of priority_queue

- `priority_queue` requires 3 template parameters
- `priority_queue<T, Container = vector<T>, Compare = less<T>>`
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
  - *Second* is the container (for now, just don't think about it)
  - *Third* is the class for comparator (the class that we use to compare)
    - This one is default to `less<T>`

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
  less<int> x;     // comparator that checks a < b
  greater<int> y; // checks a > b

  int a = 10;
```

```
    int b = 3;
    cout << x(a,b) << endl;   // 0
    cout << y(a,b) << endl;   // 1
}
```

Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
  - Either make default ordering (overload <) in the custom class
  - Or use custom comparator when declare
- For set, the declaration is `set<T, Compare = less<T>>`
- For map, the declaration is `map<Key, T, Compare = less<Key>>`

Assignment

- Is any of `vector<int>`, `set<int>`, `map<int,string>`, `queue<bool>`, `stack<vector<int>>` comparable?
  - For any class that is "YES", how it is ordered?
  - For example, if `vector<int>` is comparable, how `{1,2,3}` is compared to `{1,2,3,4}` or `{2,3,4}`

Answer

- `vector<int>` compare each element from the left; `{2,3,4} > {1,2,3,4} > {1,2,3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int,string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

Summary

Data Structure Summary

<table>
<tr><th>Data Structure</th><th>Pro</th><th>Cons</th><th>Remark</th></tr>
<tr><td><code>pair<T1,T2></code></td><td colspan="2">Nothings… It just a pair of two data type</td><td></td></tr>
<tr><td><code>vector<T></code></td><td>• Fast access <code>[ ]</code>
• Fast append $\mathcal{O}(1)$</td><td>• Slow find $\mathcal{O}(N)$
• Slow insert, Slow Erase $\mathcal{O}(N)$</td><td></td></tr>
<tr><td><code>set<T></code></td><td rowspan="2">• Fast find $\mathcal{O}(\log(N))$
• Item is sorted</td><td rowspan="2">• Slower to just append data than vector, stack, queue $\mathcal{O}(\log(N))$
• Iterate is also slow
• Takes lots of memory $\mathcal{O}(\log(N))$</td><td>Require comparator</td></tr>
<tr><td><code>map<Key,T></code></td><td>• Associative data type
• Also require comparator of Key_Type</td></tr>
<tr><td><code>stack<T></code></td><td rowspan="2">• Very fast push, pop $\mathcal{O}(1)$</td><td rowspan="2">• Very limited functionality but has special uses</td><td rowspan="3">• No iterator
• Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value)
• pq requires comparator</td></tr>
<tr><td><code>queue<T></code></td></tr>
<tr><td><code>priority_queue<T></code></td><td>• Fast get max
• Fast delete max
• Data is sorted
• Memory efficient</td><td>• Slower to just append data than vector, stack, queue
• Very limited functionality</td></tr>
</table>

More Data Structure

- C++ has more data structure not really covered right now
  - `list` is a vector with faster `input` / `erase` but does not have fast access
  - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
  - `deque` is a queue that can push, pop at both ends

- `multiset`, `multimap` are set and map that allows duplicated entries

Lecture 8

Create Your Own CP::pair

Intro

- We start writing our own data structure
  - In our own *namespace* (CP)
- We start with *pair*
  - Good introduction on writing a class
  - Some C++ feature we need to use
    - const
    - pass-by-value, pass-by-ref
    - Header file

What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to stop in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
  - Include lots of special function such as *constructor*, *destructor*, *overloading*

CP:pair version 0.1 (minimal version)

- Templating
  - Parameter of a code
- No member functions
  - When we declare a pair, `T1` and `T2` must be declared as a type

```
namespace CP {
  template <typename T1, typename T2>
  class pair{
    public:
      T1 first;
      T2 second;
  };
}

int main() {
  CP::pair<int,string> p1;  // In p1, T1 is int, T2 is string
  CP::pair<bool,int> p2;    // In p2, T1 is bool, T2 is int
}
```

Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
```

```
namespace CP {
  template <typename T1, typename T2>
  class pair{
    public:
      T1 first;
      T2 second;
  };
}

int main() {
  CP::pair<int,string> p1, p2;  // default ctor
  p1.first = 20; p1.second = "somchai";
  CP::pair<int,string> a(p1);   // copy ctor
  p2 = p1;

  cout << p2.first << "," << p2.second << endl;
}

/*
if (p1 == p2) { // won't compile
  cout << "yes" << endl;
}

if (p1 < a) { // won't compile
  cout << "yes" << endl;
}
*/
```

CP::pair version 0.2 (comparator overload)

```
#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
  template <typename T1, typename T2>
  class pair{
    public:
      T1 first;
      T2 second;

      // operator
      bool operator==(const pair<T1,T2> &other) {
        return (first == other.first && second == other.second);
      }

      bool operator<(const pair<T1,T2>& other) const {  // operator< must be const
        return ((first < other.first) ||
                (first == other.first && second < other.second));
      }
  };
}

int main() {
  CP::pair<int,string> p1, p2;  // default ctor
  p1.first = 20; p1.second = "somchai";
```

```
  CP::pair<int,string> a(p1);   // copy ctor
  p2 = p1;
  cout << p2.first << "," << p2.second << endl;  // 20, somchai

  if (p1 == p2) {cout << "yes" << endl;}  // yes
  if (p1 < a) {cout << "yes" << endl;}
  set<CP::pair<int,int>> s;
  s.insert({1,2});
  // Now we can compare and use less than (and also set, map, priority queue)
  cout << s.begin()->second << endl;  // 2
}
```

Const Member

Const Parameter in Function

```
void test(int &x, const int& y) {
  x++;                // Okay; doable
  cout << y << endl;  // Okay, we only read y
  for (int i=0; i<y; i++) { // Okay, too
    cout << i << endl;
  }
  y--;  // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword const as a prefix
- Const parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
  - Especially when we pass-by-reference

Difference of Pass-by-reference && Pass-by-value

Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
  - SLOWER!!! Because we have to copy

Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
  - Modify the *parameter* also modify the *argument's* variable
- Faster

Const Member Function

```
class ccc {
public:
  int a,b;

  void inspect() const {  // This function promises NOT to change anything
    if (a < b) cout << "yes" << endl;  // Okay

    // b += 20;  // <-- NOT OKAY
  }

  void mutate() {  // This function might change something
    if (a < b) a += 10;  // Okay
```

```
  }
};

void test2(ccc& changeable, const ccc& unchangeable) {
  changeable.inspect();
  changeable.mutate();
  unchangeable.inspect();
  unchangeable.mutate();  // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword `const` after the function declaration
- Const member function cannot modify any member data
  - Also cannot call any other function that is not const
- Why? So that we know that the function does not modify its data

Constructor & List Initialization

Custom Constructor

- In STL spec of std::pair, it has custom constructor called *intialization constructor*

```
namespace CP {
  template <typename T1, typename T2>
  class pair{
    public:
      T1 first;
      T2 second;

      // Custom constructor, using initialization list
      pair(const T1 &a, const T2 &b) {
        first = a;
        second = b;
      }
      // When we have a constructor, a default constructor is not auto-generated

      // Operator
      bool operator==(const pair<T1,T2> &other) {...}
      bool operator<(const pair<T1,T2>& other) {...}
  };
}

int main() {
  CP::pair<int, bool> p(10,false);
  CP::pair<string, int> q("abc",42), r("",0);

  cout << (q < r) << endl;  // 0
  priority_queue<CP::pair<string,int>> pq;
  pq.push(r);
  pq.push(q);
  cout << pq.top().first << endl;  // abc

  CP::pair<string, int> x(q);
  CP::pair<string, int> y = x;

  // All below cannot be compiled
  // CP::pair<string, int> w;
```

```
  // vector<CP::pair<int, int>> v(10);
}
```

Initialization List

- Instead of writing a code to assign a value to each member, we can use *intialization list*
  - A little bit shorter code
  - Also little bit faster
  - Only way to init *const* member

```
namespace CP {
  template <typename T1, typename T2>
  class pair {
    public:
      T1 first;
      T2 second;

      // custom constructor, using initialization list
      pair(const T1 &a, const T2 &b) : first(a), second(b) { }
  };
}
```

Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```
namespace CP {
  template <typename T1, typename T2>
  class pair {
    T1 first;
    T2 second;

    // Constructor
    pair() : first(), second() { }
    pair(const T1 &a, const T2 &b) : first(a), second(b) { }
  };
}
```

Include & Header file

Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
  - It is better NOT TO copy our data structure code into each each file
- Rather, put our code in a file and include it where it is needed
  - Better if we want to change it
  - Better compilation
- Introducing ".h" files

C++ header file (.h) and #include

- To put a content of one file into another file, we use `#include "filename"` keyword
- C++ will simply put the content of *filename* into where we `#include` it
- `#include` has more benefit
  - Separation of *declaration* (what it is) and *definition* (how it works)
  - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

Problem with include

a.h

```
class a {
  int m1, m2;
};
```

b.h

```
#include "a.h"

class b {
  a x;
};
```

c.h

```
#include "a.h"

class c {
  a y;
};
```

main.cpp

```
#include "b.h"
#include "c.h"

int main() {
  b b1;
  c c1;
}
```

main.cpp (expanded)

```
class a {
  int m1, m2;
};

class b {
  a x;
};

// There are multiple copies of class a
class a {
  int m1, m2;
};

class c {
  a y;
};

int main() {
  b b1;
  c c1;
}
```

Fix problem

a.h

```
#ifndef A_H
#define A_H
class a {
  int m1, m2;
};
#endif
```

b.h

```
#include "a.h"

class b {
  a x;
};
```

c.h

```
#include "a.h"

class c {
  a y;
};
```

main.cpp (expanded)

```
#ifndef A_H
#define A_H
class a {
  int m1, m2;
};
#endif

class b {
  a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//   int m1, m2;
// };
#endif

class c {
  a y;
```

```
};

int main() {
  b b1;
  c c1;
}
```

Summary

Major

- Templating: allow user to write a code that works with different data type
- Constructor: a function called when we create a variable
  - Default constructor
- Operator overloading: for less-than and equality
- pass-by-ref vs pass-by-value

Minor

- Header file
- const
- List initialization

Exercise (no grader)

1. Modify `operator<` so that it compare *second* before *first*
2. Modify `operator<` so that when we call `sort(v.begin(),v.end())` where `w` is a vector of our pair, it is sorted from *Max* to *Min*
3. Write `operator!=` and `operator>=`

Final version CP.h

```
#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

namespace CP {
  template <typename T1, typename T2>
  class pair {
    public:
      T1 first;
      T2 second;

      pair() : first(), second() {}

      pair(const T1 &a, const T2 &b) : first(a), second(b) { }

      bool operator==(const pair<T1,T2> &other) {
        return (first == other.first && second == other.second);
      }


      // 1.
      bool operator<(const pair<T1,T2>& other) const {
        return ((second < other.second) ||
                (second == other.second && first < other.first));
      }

      // 2.
      bool operator<(const pair<T1,T2>& other) const {
        return ((first > other.first) ||
```

```
                (first == other.first && second > other.second));
      }

      // 3.
      bool operator!=(const pair<T1,T2>& other) const {
        return !(*this == other);
      }

      bool operator>=(const pair<T1,T2>& other) const {
        return !(*this < other);
      }

  };
}
#endif
```

Lecture 9

CP::vector

Our first "real" data structure

Intro

- Now we will create more complex data structure `CP::vector`
- It can store variable length array
  - Implemented as a dynamic array
- Can be accessed by `operator[]`
  - Additional operator to be overloaded
- We also have to create our own iterator
  - Implemented as a pointer

Key Idea

- Vector stored 3 things (3 member data)
  - mData: A dynamic array, large enough space to store current data and might have reserved space
  - mSize: Number of data stored
  - mCap: Size of the dynamic array (maybe large than mSize)
- If the dynamic array is full and more data is being added, we create a new dynamic array and relocate data to the new array
  - This is called expand
  - Each expansion takes very long time
- Dilemma
  - Large reserve = less often relocation but use more memory
  - Small reserve = more frequent relocation but less memory

Example

```
mSize  mCap       mData
  4      5     [10, 3, 1, 5, ]

push_back(9)

  5      5     [10, 3, 1, 5, 9]

push_back(4)
```

```
6     10    [10, 3, 1, 5, 9, 4, , , , ]
```

How much reserve should we have?

- Whenever we need to relocate, we double the capacity that we currently have
- If we start with a vector with zero size and continuously add data one by one, for example by `push_back`
  - Each expansion will take time equal to the number of data when we relocate (this is very slow)
- By doubling the size every time we expand, we can show that on average, each addition of a data (such as `push_back`, `insert`) takes constant time!!!

Pointer

How to implement a dynamic array (and also iterator)

Pointer & Memory

- Each variable is some block in computer memory
  - Programming language just map our variable name to that block of memory
  - Programming language works with the address of that block
- Pointer variable is a variable that stor address of memory
  - Pointer needs type i.e., address of int is not the same as address of bool
  - We can use operator `&` to ask for the address of a variable
  - We can use operator `*` to ask for the data of an address

Example

```
int x, y;  // this is normal variable x and y
int *a;    // this is int pointer variable a
int *b;    // another int pointer

x = 10;    // say x is at address 3267
a = &x;    // a points to x
b = a;     // b also points to x
*b = 30;   // change the address value to 30
y = *b;    // set value of y to 30 (hard copy)

cout << &x << endl;  // 0x7ffee22885ac
cout << &y << endl;  // 0x7ffee22885a8  +4
cout << &a << endl;  // 0x7ffee22885a0  +8
cout << &b << endl;  // 0x7ffee2288598  +8
cout << sizeof(int) << endl;   // 4 bytes
cout << sizeof(*int) << endl;  // 8 bytes
```

Pointer Arithmetic

- Pointer can be added, subtracted by integer
  - It moves the address by the size of the type of the pointer
  - For example, when `X` is an int (which is 4 bytes) `X + 10` result in an address 40 bytes away from `X`
- Two pointers of the same type can be subtracted
  - The result is the address difference divided by size of the type of the pointer

```
int main() {
  bool x, y, z;
  x = 1; y = 2; z = 3;
  bool *a, *b;

  a = &x;
  b = a+2;
```

```
  cout << "&x = " << &x << endl;  // &x = 0x6afed4
  cout << "&y = " << &y << endl;  // &y = 0x6afed8
  cout << "&z = " << &z << endl;  // &z = 0x6afedc
  cout << " a = " << a  << endl;  //  a = 0x6afed4
  cout << " b = " << b  << endl;  //  b = 0x6afedc
  cout << b-a << endl;            // 2
}
```

Dynamic Array

- Dynamic Array variable of type `T` is a pointer to the starting address of consecutive block of type `T`
- Let `A` be a dynamic array of int
  - `A[x]` refer to the x^{th} block starting from `A`
- Static array works in the same way, that's why accessing `A[x]` is very fast for an array
  - It just refers to the address `A[x]` is `*(A + x * size_of(T))`

new and delete operator

- For a datatype `T`, `new T` *allocates* a block with a size of `T` and then call a *constructor* of `T`, return the address of that memory
- For a datatype `T`, `new T[n]` *allocates* `n` blocks of `T` and then call a *constructor* of `T` in each block, return the address of the first block
  - For example, we use `new int[10]` to create a dynamic int array of 10 elements
- For a pointer `X`, `delete X` calls the *destructor* and then *de-allocates* memory pointed by `X`
- For a dynamic array `X`, `delete [ ]` calls the *destructor* of all blocks in the dynamic array and *de-allocates* the memory allocated by `X`

Example

```
class test {
public:
  // constructor
  test() : data() {cout << "created" << endl;}
  // destructor
  ~test() {cout << data << " destroyed " << endl;}
  int data;
};

int main() {
  test *a, *b;    // pointers of test

  a = new test;   // created new test pointed by a
  a->data = 10;   // (*a).data = 10
  cout << a->data << endl;  // 10
  delete a;       // destroyed

  b = new test[4];   // created 4x
  b[0].data = 10;
  b[1].data = 20;
  b[2].data = 30;
  b[3].data = 40;
  delete [] b;       // deleted 4x
}
```

Memory Leak

- For everything, this is created by `new`, we must call `delete` on it
- If you do not, that memory is not deleted until all memory is used up

```
#include <iostream>
using namespace std;

void leaked() {
  int *a;
  a = new int[2000];
}
int main() {
  for (int i=0; i<1000000; i++) {
    int *a;
    cout << i << endl;
    leaked();
  }
}
// terminate called after throwing an instance of 'std::bad_alloc'
```

Smart Pointer (NOT A SUBJECT OF THIS COURSE)

- A better way is to use C++ smart pointer `shared_ptr(T)`
- Smart pointer is a pointer that can delete itself when it go out of scope
- Similar concept to Java Garbage Collection
- Still possible to have memory leak

vector.h

Version 0.1

- Start with vector that can do push_back, pop_back, and []
- Also with custom constructor

```
namespace CP {
template <typename T>
  class vector {
    protected:
      T *mData;
      size_t mCap;
      size_t mSize;

      void rangeCheck(int n) {...}
      void expand(size_t capacity) {...}
      void ensureCapacity(size_t capacity) {...}
    public:
      vector() {...}
      vector(size_t capacity) {...}
      ~vector() {...}
      // access
      T& at(in index) {...}
      T& operator[](int index) {...}
      // modifier
      void push_back(const T& element) {...}
      void pop_back() {...}
  };
}
```

Basic Constructor

```
template <typename T>
class vector {
```

```
  protected:
    T *mData;
    size_t mCap;
    size_t mSize;

  public:
    vector() {
      int cap = 1;
      mData = new T[cap]();
      mCap = cap;
      mSize = 0;
    }

    vector(size_t cap) {
      mData = new T[cap]();
      mCap = cap;
      mSize = cap;
    }
```

```
vector<int> v;                      vector<int> w(5);

mSize  mCap  mData                  mSize  mCap     mData
  0     1     [ ]                     5     5    [0,0,0,0,0]
```

Destructor

- Since we `new mData`, we have to `delete` it
  - Or face a memory leak problem

```
    ~vector() {
      delete [] mData;
    }
};
```

Object Life Cycle

- Normal object
  - Object is created (constructor called) when declared
  - Object is destroyed (destructor called) when go out of scope
- Object created by new (both `new T` or `new T[]`) Object is created when `new` Object is destroyed when `delete`

```
class test {
public:
  // constructor
  test() : data() {cout << "created" << endl;}
  // destructor
  ~test() {cout << data << " destroyed " << endl;}
  int data;
};

int main() {
  cout << "-- Life cycle --" << endl;
  cout << "- normal cycle -" << endl;
  test u;
  u.data = 99;
  for (int i=0; i<5; i++) {
    test t;
```

```
    t.data = i*10;
  }
}
```

Output

```
-- Life cycle --
- normal cycle -
created       <-- u
created
0 destroyed
created
10 destroyed
created
20 destroyed
created
30 destroyed
created
40 destroyed
99 destroyed  <-- u
```

Accessing Data

- The return type is `T&` which is a reference
- Same deal as pass-by-refernce, this is called return-by-reference
  - So we can do `v[i] = 30` or `v[i]++`
- What is returned is actually that variable
- Also notice the difference between `at()` and `operator[]`

```
template <typename T>
class vector {
  protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {
      if (n < 0 || (size_t)n >= mSize) {
        throw std::out_of_range("index out of range");
      }
    }

  public:
    T& at(int index) {
      rangeCheck(index);
      return mData[index];
    }

    T& operator[](int index) {
      return mData[index];  // return the variable
    }
};
```

Add, remove data

- push_back first check of we have reserved space
  - If not, we expand
- Then, the data is put to mData[mSize]

- Removing Data is done by just reduce the size

```
template <typename T>
class vector {
  protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    void expand(size_t capacity) {
      T *arr = new T[capacity](); // create a new dynamic array
      for (size_t i = 0; i < mySize; i++) {
        arr[i] = mData[i]; // move all data
      }
      delete [] mData;
      mData = arr; // delete old data and point to new one
      mCap = capacity;
    }
    void ensureCapacity(size_t capacity) {
      if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap; // double the size
        expand(s);
      }
    }
  public:
    void push_back(const T& element) {
      ensureCapacity(mSize+1);
      mData[mSize++] = element; // add size and put element
    }
    void pop_back() {
      mSize--;
    }
};
```

Vector Constructors

Problem of v0.1

- Copy constructor and assignment operator is incorrect
  - It is auto generate to copy all variables (but not the data it points to)
- Rule of three in c++
  - Consider destructor, copy constructor, assignment operator
  - If any of them is written in the code, we mostly need all of them
- Since c++11, it's rule of four an a half

```
int main() {
  CP::vector<int> w(5);

  for (int i=0; i<5; i++) w[i] = i*10;
  CP::vector<int> x(w); // this creates shallow copy of w
  CP::vector<int> y = w;
  x[3] = -1; // changes both w and y
  cout << y[3] << endl; // -1
  cout << w[3] << endl; // -1
}
```

v0.2, add small access functions

- `empty` and `size` also exists in other data structure

- size_t is non-negative integer type

```
bool empty() const {
  return mSize == 0;
}

size_t size() const {
  return mSize;
}

size_t capacity() const {
  return mCap;
}
```

v0.2 adding copy constructor & assignment operator

```
// copy constructor
vector(const vector<T>& a) {
  mData = new T[a.capacity()]();
  mCap = a.capacity();  // copies only value
  mSize = a.size();
  for (size_t i=0; i<a.size(); i++) {
    mData[i] = a[i];
  }
}

// copy assignment operator
vector<T>& operator=(vector<T> &other) {
  // protect against self-destruct
  if (mData != other.mData) {
    // delete current data
    delete [] mData;
    // copy the new data
    mData = new T[other.capacity()]();
    mCap = other.capacity();
    mSize = other.size();
    for (size_t i=0; i<a.size(); i++) {
      mData[i] = a[i];
    }
  }
}
```

Copy-and-swap

- Utilize written copy-constructor nd destructor
- Shorter code

```
// copy assignment operator using copy-and-swap idiom
vector<T>& operator=(vector<T> other) {  // notice the pass-by-value!!!
  // other is copy-constructed which will be destruct at the end of this scope
  // we swap the content of this class to the other class and let it be destructed
  using std::swap;
  swap(this->mSize, other.mSize);
  swap(this->mCap, other.mCap);
  swap(this->mData, other.mData);
  return *this;
}
```

Iterator, Insert, Erase, & Analysis

v0.3 Iterator and typedef keyword

```
template <typename T>
class vector {

  protected:
    T *mData;
    size_t mCap;
    size_t mSize;
  protected:
    typedef T* iterator;  // let iterator be type name of T pointer

    // iterator
    iterator begin() {
      return &mData[0];
    }

    iterator end() {
      return begin()+mSize;
    }
}
```

- See that pointer works just like how `std::vector::iterator` works
- In fact, iterator is actually a pointer
- `typedef` keyword allows us to map a type name
  - `CP::vector<int>::iterator` is `int*`
  - `CP::vector<bool>::iterator` is `bool*`

insert

- push_back actually call `insert(end(), element)`
- Question: why we need pos?

```
iterator insert(iterator it, const T& element) {
  size_t pos = it - begin();
  ensureCapacity(mSize+1);
  for (size_t i = mSize; i > pos; i--) {
    mData[i] = mData[i-1];
  }
  mData[pos] = element;
  mSize++;
  return begin()+pos;
}

void push_back(const T& element) {
  insert(end(), element);
}
```

erase

- See that both `insert` and `erase` also change `mSize`

```
void erase(iterator it) {
  while((it+1)!=end()) {
    *it = *(it+1);
    it++;
  }
```

```
  mSize--;
}
```

Exercise

- Read the following function and see how it works in `vector.h`
  - `resize`, `clear`
  - non-stl function
    - `insert_by_pos`
    - `erase_by_pos`
    - `erase_by_value`
    - `constains`
    - `index_of`
- Read in here

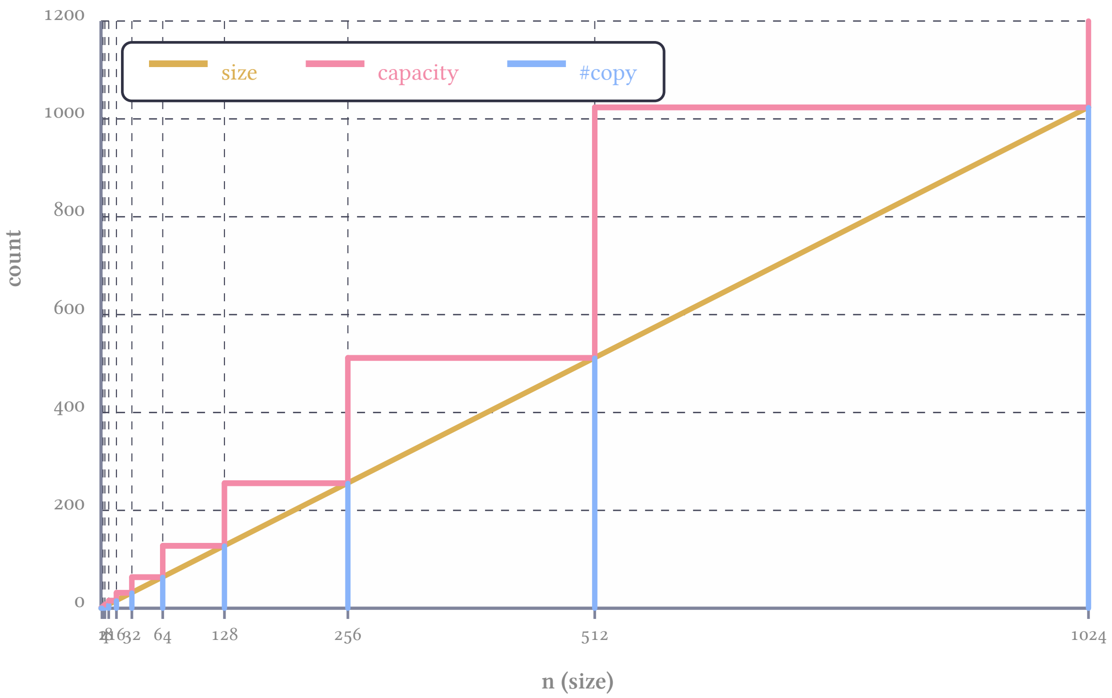
Will do this part later

Analysis of how many data is copied by push_back

- When full, `push_back` have to move all data to a new dynamic array
- `ensureCapacity` double the size

Size and Capa & Copy Count

- How much copy we need?



Lecture 10

Stack

Intro

- Now we will create less complex data structure `CP::stack`
- Just like a vector without `iterator, insert, erase, resize, at end operator[]`

  - Add `top()` which is just a shorthand of looking at the last element
- That's it, really

Key Idea

- The data is stored in the same way as a vector
  - The first element of `mData` is the bottom of stack while the last element is the top of the stack
- We just take `vector.h` and remove unnecessary function

```
|   |   stack
|   |
| A |   mSize  mCap     mData
| X |     4      5    [A, B, X, A]
| B |
| A |
|___|
```

stack.h

```
namespace CP {
  template <typename T>
  class stack {
    protected:
      T *mData;
      size_t mCap;
      size_t mSize;
      void expand(size_t capacity) {...}
      void ensureCapacity(size_t capacity) {...}
    public:
      // constructor
      stack(const stack<T>& a) {...}
      stack() {...}
      stack<T>& operator= {...}
      ~stack() {...}
      // capacity operation
      bool empty() const {...}
      size_t size() const {...}
      // access
      const T& top() const {
        return mData[mSize-1];
      }
      // modifier
      void push(const T& operator) {...}  // This is push_back
      void pop() {...}  // This is pop_back
  };
}
```

Speed of each operation

- All read operation always take constant time
  - `size()`, `top()` simply return something that is directly accessible
- All modify operation also take constant time
  - `push()` is constant on average (same as `push_back` of vector)
  - `pop()` is always constant

Stack by Vector

- Instead of writing our own function, there is another way to write a stack
- We simply use vector as our sole data member

- Benefit: code reuse
- Drawback: almost none except that we need one more layer of function call

```
namespace CP {
  template <typename T>
  class stack {
    protected:
      vector<T> v;
    public:
      // default constructor
      stack() : v() { }
      // capacity function
      bool empty() const { return v.empty(); }
      size_t size() const { return v.size(); }
      // access
      const T& top() const { return v.back(); }
      // modifier
      void push(const T& element) { v.push_back(element); }
      void pop() { v.pop_back(); }
  };
}
```

Lecture 11

CP::queue

Will the circle be unbroken?

Intro

- Queue, unlike stack, require more sophisticated technique to achieve fast performance
- We start by writing a simple class that just work (slowly)
- Then we try to improve it

Key Idea

- Just like stack, we will use the same format as vector, using dynamic array to store data
- However we have to somehow manage how we works with `front()` and `back()` of the queue

v0.1 simple implementation of the queue

- To illustrate this idea, we will use a vector as our data member
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[0]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `v.erase(v.begin())`, this is slow (always proportional to `v.size()`)
  - Unlike `std::queue` which has very fast `pop()`

```
namespace CP {
  template <typename T>
  class queue {
    protected:
      std::vector<T> v;
    public:
      // default constructor
      queue() : v() { }
      // capacity function
      bool empty() const { return v.empty(); }
```

```
    size_t size() const { return v.size(); }
    // access
    const T& front() const { return v[0]; }
    const T& back() const { return v[v.size()-1]; }
    // modifier
    void push(const T& element) { v.push_back(element); }
    void pop() { v.erase(v.begin()); }
  };
}
```

v0.1 example

```
CP::queue<int> q;     q  v
                         mSize  mCap  mData
                           0     1      ---->  [ ]
q.push(10);           q  v
                         mSize  mCap  mData
                           1     1      ---->  [10]
q.push(20);           q  v
                         mSize  mCap  mData
                           2     2      ---->  [10, 20]
q.push(30);           q  v
                         mSize  mCap  mData
                           3     4      ---->  [10, 20, 30, ]
q.pop();              q  v                         /    /
                         mSize  mCap  mData       /    /
                           2     4      ---->  [20, 30, , ]
```

v0.2 faster queue

- Add more data member `mFront`, initialized as `0`
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[mFront]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `mFront++`, this is fast
  - However, we don't really remove anything when pop

```
#include <vector>

namespace CP {
  template <typename T>
  class queue {
    protected:
      std::vector<T> v;
      int mFront;
    public:
      // constructor
      queue() : v(), mFront() {}
      // capacity function
      bool empty() const { return v.empty(); }
      size_t size() const { return v.size() - mFront; }
      // access
      const T& front() const { return v[mFront]; }
      const T& back() { return v[v.size()-1]; }
      // modifier
      void push(const T& element) { v.push_back(element); }
      void pop() { mFront++; }
  };
}
```

v0.2 example

```
CP::queue<int> q;     q          v
                         mFront  mSize  mCap  mData
                           0       0     1      ---->  [ ]
q.push(10);           q          v
                         mFront  mSize  mCap  mData
                           0       1     1      ---->  [10]
q.push(20);           q          v
                         mFront  mSize  mCap  mData
                           0       2     2      ---->  [10, 20]
q.push(30);           q          v
                         mFront  mSize  mCap  mData
                           0       3     4      ---->  [10, 20, 30, ]
q.pop();              q          v
                         mFront  mSize  mCap  mData
                           1       3     4      ---->  [10, 20, 30, ]
```

Problem with v0.2

- Fast but use too many space
- Queue grows according to how many time push is called
  - regardless of how many pop is called
- The data stored in the vector can be much larger than the actual data in the queue
- Does not really work in real world

```
for (int i=0; i<1000000; i++) {
  q.push(i);
  q.pop();
}
std::cout << q.size() << std::endl;
```

Circular Queue

Final Idea

```
  mFront  mSize  mCap  mData              *
    2       2     4      ---->  [   ,   , 30, 40]
  q.push(50);
  mFront  mSize  mCap  mData              *
    2       3     4      ---->  [50,    , 30, 40]
  q.push(60);
  mFront  mSize  mCap  mData              *
    2       4     4      ---->  [50, 60, 30, 40]
  q.pop();
  mFront  mSize  mCap  mData                  *
    3       3     4      ---->  [50, 60,    , 40]
  q.push(99);
  mFront  mSize  mCap  mData                  *
    3       4     4      ---->  [50, 60, 99, 40]
  q.push(1);                        \   \   \
  mFront  mSize  mCap  mData      *  \   \   \
    0       5     4      ---->  [40, 50, 60, 99,  1,   ,   ,   ]
```

- We take v0.2 and reuse the area at the beginning of `mData`
  - Expand when necessary
  - Re-arrange when expand

Circular Queue

- We can think of `mData` to be circular
  - End of the last element of the `mData` is connected to the first element
- Consider i^{th} element
  - the next element is `(i+1) % mCap`
  - The previous element is `(i-1+mCap) % mCap`
  - Next k element is `(i+k) % mCap`

Circular Queue Implementation

queue.h

```
namespace CP {
  template <typename T>
  class queue {
    protected:
      T *mData;
      size_t mCap;
      size_t mSize;
      size_t mFront;
      void expand(size_t capacity) {...}
      void ensureCapacity(size_t capacity) {...}
    public:
      // constructor - almost the same but have to take care of mFront
      queue(const queue<T>& a) {...}
      queue() {...}
      queue<T>& operator=(queue<T> other) {...}
      ~queue() {...}
      // capacity function - same as vector
      bool empty() const {...}
      size_t size() const {...}
      // access - circular queue implementation
      const T& front() const {...}
      const T& back() const {...}
      // modifier
      void push(const T& element) {...}
      void pop() {...}
  };
}
```

Ctor, Dtor, copy

```
template <typename T>
class queue {
  protected:
    T *mData;  size_t mCap;  size_t mSize;  size_t mFront;
  public:
    // default constructor
    queue() : mData(new T[1]()), mCap(1),
              mSize(0), mFront(0) { }
    // copy constructor
    queue(const queue<T>& a) : mData(new T[a.mCap]()), mCap(a.mCap),
                               mSize(a.mSize), mFront(a.mFront) {
      for (size_t i = 0; i < a.mCap; i++) {
        mData[i] = a.mData[i];
      }
```

```
    }
    // copy assignment operator
    queue<T>& operator=(queue<T> other) {
      using std::swap;
      swap(mSize, other.mSize);
      swap(mCap, other.mCap);
      swap(mData, other.mData);
      swap(mFront, other.mFront);
      return *this;
    }
    ~queue() {
      delete [] mData;
    }
};
```

front(), back(), pop()

```
template <typename T>
class queue {
  protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
  public:
    // access
    const T& front() const {
      return mData[mFront];
    }
    const T& back() const {
      return mData[(mFront + mSize - 1) % mCap];
    }
    // modify
    void pop() {
      mFront = (mFront + 1) % mCap;
      mSize--;
    }
};
```

- `back = mFront + mSize - 1`
  - Also circular (by `% mCap`)
- `pop` = move `mFront` by 1
  - Also circular
  - Also change size

push, expand

- push add data to `(mFront + mSize) % mCap`
  - The space just after `back()`
- Expand re-pack the `mData` so that `mFront` is 0
- `ensureCapacity` is the same

```
template <typename T>
class queue {
  protected:
    T *mData;
    size_t mCap;
    size_t mSize;
```

```
    size_t mFront;
    void expand(size_t capacity) {
      T *arr = new T[capacity]();
      for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[(mFront + i) % mCap];
      }
      delete [] mData;
      mData = arr;
      mCap = capacity;
      mFront = 0;
    }
    void ensureCapacity(size_t capacity) {
      if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
        expand(s);
      }
    }
  public:
    void push(const T& element) {
      ensureCapacity(mSize+1);
      mData[(mFront + mSize) % mCap] = element;
      mSize++;
    }
};
```

Analysis

- All access, modification is fast (constant time)
- Space is re-used
  - It is not shrunk when `mSize` reduce
  - Space is not more than double of maximum `mSize` during its lifetime

Queue Exercise

- We implement circular queue by maintain `mFront` and use circular logic (`% mCap`) to calculate the position of back of the queue
  - Can we maintain `mBack` instead?
  - Can we maintain both `mFront` and `mBack` but not `mSize`?
- How about `mCap`, if we know `mFront`, `mSize`, `mBack`, can we calculate `mCap`?

mFront	mSize	mBack	front()	back()	size()
YES	YES	No	v[mFront]	v[(mFront + mSize) % mCap]	mSize()
No	YES	YES	????	v[mBack]	mSize
YES	No	YES	v[mFront]	v[mBack]	????

Answer

- Use `front() = v[(mCap + mBack - mSize) % mCap]`
- Use `size() = (mCap + mBack - mFront) % mCap`, but this will break when `mBack == mFront` because it can't determine whether the queue is empty or full.
- Impossible, `mCap` can be scaled to any size larger than the current elements.

Now, meet deque

- Can you modify queue to include
  - `push_front()`, add to the front of the queue
  - `pop_back()`, remove from back of the queue
- All operation should still be constant time

Answer

```
void push_front(const T& element) {
  ensureCapacity(mSize+1);
  mFront = (mCap + mFront - 1) % mCap;
  mData[mFront] = element;
  mSize++;
}
void pop_back() {
  mSize--;
}
```

Lecture 12

Complexity Analysis

How fast is our code?

Overview

- Introducing a measurement of efficiency of a program
- Why we need measurement
- How we measure
  - How to analyze our code to get a measurement
- What is measurement
  - Asymtotic Notation

Key Idea

- We need a useful way to describe effiency of our code
  - Useful = able to be easily use to predict how much resource (time / memory) that our program will need
  - Useful = not overly complex in analysis
  - Need to balance between usefulness and complexity
- Ultimately, we introduce a class of efficiency that says how our code use resource with respect to size of data
  - Focus on on growth of resource usage

Preview

```
int find_max(vector<int> v) {
  int m = v[0];
  for (size_t i = 0; i < v.size(); i++)
    if (v[i] > m)
      m = v[i];
  return m;
}
```

- This code takes time directly proportional to the size of the data
  - Size N takes time T
  - Size $5N$ should take time $5T$

```
int count_pair_sum(vector<int> v, int k) {
  int count = 0;
  for (size_t i = 0; i < v.size(); i++)
    for (size_t j = 0; j < v.size(); j++)
      if (i != j && v[i] + v[j] == k)
        count++;
  return count/2;
}
```

- This code takes time directly proportional to the square of the size of the data
  - Size N takes time T
  - Size $5N$ should take time $25T$

Why don't we use real world clock?

- Ultimately, we want to know how long our program takes to do each operation

  - Use in design, how much resource we need
  - Help use choose appropriate data structure
- Real world clock measurement is possible but has many drawback
  - System dependency
  - Too complex (we have to build our system)
  - Too specific

Dependency of System

- Same program in different system = different time
  - CPU
  - RAM
  - Compiler
  - Operating parameter
    - Heat? Other running program?

How to measure

- Counting instruction
  - Depend on code only
- Dependency on size of data
  - Focus on large data

```
int count_pair_sum(vector<int> v, int k) {
  int count = 0;                              // 1
  for (size_t i = 0; i < v.size(); i++)       // 2 + n + n   ( n = v.size() )
    for (size_t j = 0; j < v.size(); j++)     // (2 + n + n) * n
      if (i != j && v[i] + v[j] == k)         // 2 * n * n
        count++;                              // 1 * ????  ( <= (d) )
  return count/2;                             // 1
}

// Total = 1 + 2+n+n + (2+n+n)n + 2n^2 + n^2 + 1
//       = 4 + 4n + 5n^2
```

Time per instruction

- In reality, different CPU instructions use different time
- Same instruction but different CPU also use different number of cycle
- However, we just ignore it
  - For now

Focusing on large data

- In many cases, the size of the data we are working with will affect the time our code use
- Large data usually mean longer time
- What matter is when the data is large

Growth rate simplifies analysis

```
int count_pair_sum(vector<int> v, int k) {
  int count = 0;                              //      i=0: 1 + n-0 + n-1
  for (size_t i = 0; i < v.size(); i++)       //      i=1: 1 + n-1 + n-2
    for (size_t j = i+1; j < v.size(); j++)   // <-- i=2: 1 + n-2 + n-3
      if (v[i] + v[j] == k)                   //      . . .
        count++;                              //      i=n-1: 1 + 1 + 0
  return count;                               //      sum = Σ(1+2n-2i-1) = n^2 + n
}
```

```
// Total = 1 + 2+n+n + n^2+n + n^2 + n^2 + 1
//       = 4 + 3n + 3n^2
```

Small Detail

n	$5n^2 + 4n + 4$		$3n^2 + 3n + 4$		n^2	
	raw	growth	raw	growth	raw	growth
10	544		334		100	
20	2084	3.830 88	1264	3.784 43	400	4
40	8164	3.917 47	4924	3.895 57	1600	4
80	32 324	3.959 33	19 444	3.948 82	6400	4
160	128 644	3.979 83	77 284	3.9747	25 600	4
320	513 284	3.989 96	308 164	3.987 42	102 400	4
640	2 050 564	3.994 99	1 230 724	3.993 73	409 600	4
1280	8 197 124	3.9975	4 919 044	3.996 87	1 638 400	4
2560	32 778 244	3.998 75	19 668 484	3.998 44	6 553 600	4
5120	1.31×10^8	3.999 37	78 658 564	3.999 22	26 214 400	4
10 240	5.24×10^8	3.999 69	3.15×10^8	3.999 61	1.05×10^8	4
20 480	2.1×10^9	3.999 84	1.26×10^9	3.9998	4.19×10^8	4

When counting instruction, it is usually OK to focus on most executed line

Measurement by Growth Rate

What?

- Growth rate = how much resource usage growth with respect to change of input
  - Resource usage = number of instruction used
  - Input = size of data
- Emphasizes long term trend

Why?

- System independent
  - The result can be used to predict behavior on any system
- Focus on change of resource usage with respect to size of input
- Can regard small detail
  - Simple to calculate
  - Applicable in real world

Asymtotic Notation

Classification of growth rate

Overview

- Formally, it is a set of function having the growth rate related to something
- The definition focus on growth of the function while disregard small detail
- Also provide some workaround on dependency of value of input

What is?

- A kind notation written as $O(f(n))$
  - O can be one of $\mathcal{O}, \Theta, \Omega, o, \omega$
  - $f(n)$ is some expression
- Example $\mathcal{O}(n)$ or $\Theta(n^2 + 3)$ or $\omega(n^2 \log(n))$
- Usage
  - "This code is $\mathcal{O}(n)$" (read as Big-Oh of n)
  - "This function takes time in $\Theta(\log(n))$" (read as Big-theta of log n)

- "Time complexity of this program is $\mathcal{O}(n^2)$"

Meaning

- "A is $\Theta(f(x))$" means the growth rate of A is equal to the growth rate of $f(x)$
- "B is $\mathcal{O}(f(x))$" means the growth rate of B is less than or equal to the growth rate of $f(x)$
- For Ω, o, ω, (Big-Omega, little-oh, little-omega), we won't use it for now but the meaning is similar (which are more than or equal, less than, more than, respectively)
- Conventionally, we usually use N for the size of the data

Usage

- Let $f(n)$ be the number of instruction needed by code A when the size of data is n
- We will calculate asymtotic notation that $f(n)$ is a member of
  - Find $g(n)$ such that $f(n)$ is $\mathcal{O}(g(n))$ or $\Theta(g(n))$ (or $\Omega(g(n))$ or ...)
- Let's say we have analyze that $f(n)$ is $\mathcal{O}(g(n))$
  - We now understand that the growth rate of instruction required by n grows slower or the same as how $g(n)$ grow

Comparing growth rate of f(n) and g(n)

- The relation of growth rate of $f(n)$ and $g(n)$ depends on the value of $f(n)/g(n)$ when $n \to \infty$

$$\lim_{n\to\infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & \text{if } f(n) \text{ grows slower than } g(n) \\ c & \text{if } f(n) \text{ grows similarly to } g(n) \\ \infty & \text{if } f(n) \text{ grows faster than } g(n) \end{cases}$$

Example

- $f(n) = 4 + 3n + 4n^2$
- $g(n) = n^2$

$$\lim_{n\to\infty} \frac{4n^2 + 3n + 4}{n^2} = \lim_{n\to\infty} \left(4 + \frac{3}{n} + \frac{4}{n^2}\right) = 4$$

- Hence $f(n)$ grows similarly to $g(n)$
  - Therefore, $f(n) = \Theta(n^2)$

Another Example

- $f(n) = 0.00005n^2$
- $g(n) = 100000n$

$$\lim_{n\to\infty} \frac{0.00005n^2}{100000n} = \lim_{n\to\infty} 10^{-10}n = \infty$$

- Hence $f(n)$ grows faster than $g(n)$ (also means $g(n)$ grows slower than $f(n)$)
  - Therefore $g(n) = \mathcal{O}(n^2)$
  - and $f(n) = \Omega(10000n)$

Dependency on the value of input

- Consider `vector::insert(iterator it, T value)`
- The time it takes depends on both the size of the vector and the value of it
  - Larger size $\longrightarrow$ more time
  - Closer to `end()` $\longrightarrow$ less time

Big-O describes upper bound

- `vector::insert` is $\mathcal{O}(n)$
  - Its growth rate does not exceed n
  - There are cases that it may grow less than n (insert at `end()`)

- This is very useful in real world
  - Knowing maximum load
  - Not overly complex in analysis

Big-O Example

- Find is $\mathcal{O}(n)$
  - At worse, it can't find value and `a` & `b` points to `begin()` & `end()`
    - This case, find growth rate is n
  - At best, it always find value at the first position (a)
    - This case, find growth as 1

```
bool find(iterator a, iterator b, T value) {
  while (a < b) {
    if (*a == value)
      return true;
    a++;
  }
  return false;
}
```

L'Hôpital's Rule

$$\lim_{n\to\infty} \frac{f(x)}{g(x)} = \lim_{n\to\infty} \frac{f'(x)}{g'(x)}$$

If $f(x)$, $g(x)$ must be diffable, $g(x)$ is non-zero, $\lim_{n\to\infty} f'/g'$ exists

- Can help
- $F(n) = \log n$
- $g(n) = n^{0.5}$

$$\begin{aligned}
\lim_{n\to\infty} \frac{\log(n)}{\sqrt{n}} &= \lim_{n\to\infty} \frac{\ln(n)}{\ln(10)\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n\to\infty} \frac{\ln(n)}{\sqrt{n}} \\
&= \frac{1}{\ln(10)} \lim_{n\to\infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} \\
&= \frac{1}{\ln(10)} \lim_{n\to\infty} \frac{2}{\sqrt{n}} \\
&= 0
\end{aligned}$$

Exercise

- $f(n) = \lg^c n$
- $g(n) = n^k$
- We know that $c > 0, k > 0$
- Does $f(n)$ grow slower than $g(n)$?

Will do this part later

Big Theta

Big-Theta is tight bound

- `std::count` always go through entire array
- Regardless of the value in the array, it always perform `if (*first == value)`

```
size_t count(iterator first, iterator last, const T& value) {
  size_t ret = 0;                    // 1
  for (; first != last; ++first) {  // n + n
    if (*first == value)             // 1 * n
      ret++;                         // 1 * ???? ( <= (d) )
  }
  return ret;                        // 1
}

// Total = 1 + n+n + 1*n + 1*n + 1
//       = 2 + 4n
```

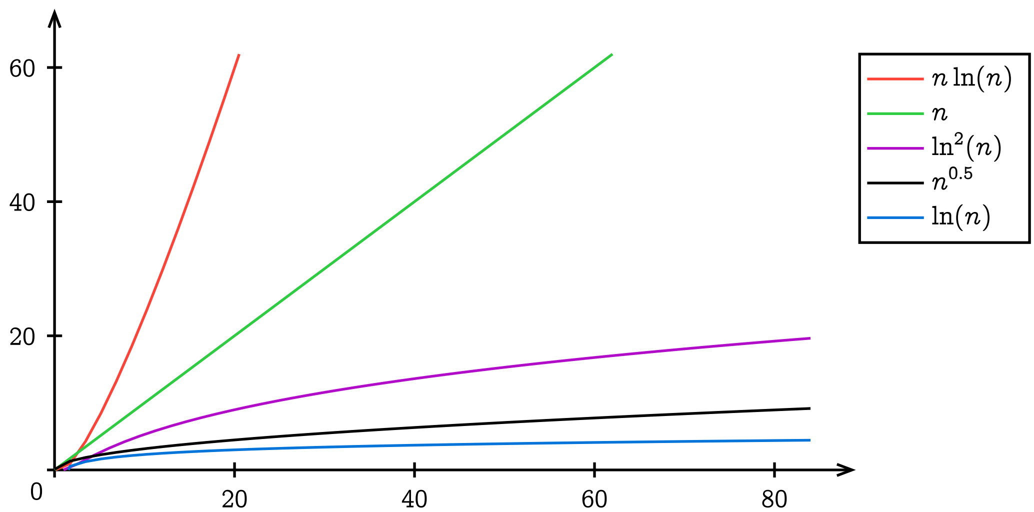
More Example

$\Theta(1)$	$\Theta(n)$	$\Theta(n^2)$
$f(n) = 5$ $f(n) = 0$ $f(n) = c$	$f(n) = n$ $f(n) = n + 3$ $f(n) = \frac{n}{1200} + 86$ $f(n) = 40000000n$	$f(n) = n^2$ $f(n) = \binom{n}{2} = \frac{n(n-1)}{2}$ $f(n) = 400n^2 + an + b$

Observation: multiplicative and addition constants in $f(n)$ can usually be ignored, since it will be disragard by lim, Degree cannot.

Well known growth rate class

$\Theta(1)$	Constant
$\Theta(\log(n))$	Logarithm
$\Theta(\log^c(n)), c \geq 1$	Polylogarithm
$\Theta(n^a), 0 < a < 1$	Sublinear
$\Theta(n)$	Linear
$\Theta(n\log(n))$	Linearithmic
$\Theta(n^2)$	Quadratic
$\Theta(n^c), c \geq 1$	Polynomial
$\Theta(c^n), c > 1$	Exponential
$\Theta(n!)$	Factorial



$\ln^2(n)$ grows slower than $n^{0.5}$. At around $n = 56000$, $\ln^2(n)$ is less than $n^{0.5}$.
x^4 grows slower than 2^n. At around $n = 16$, x^4 is less than 2^n.

Beware

- It is wrong to say that `vector::insert` is $\Theta(n)$
  - Because there is a case that it grows slower than n
- It is wrong to say that `vector::push_back` is $\Theta(n)$
  - Because there is a case that it grows slower than n
- It is ok to say that `std::count` is $\mathcal{O}(n)$
  - Because while it always grows as n, it does not grow faster than n
  - $\mathcal{O}(n)$ is upper bound
  - But it is better to say that `std::count` is $\Theta(n)$

How to analyze using asymptotic notation

1. Write a code
2. Calculate the function $f(n)$ that counts the number of instruction of the code when the data is size n
   - Usually, just focus on most executed line
3. Find $g(n)$ and a notation X such that $f(n)$ is $X(g(n))$
   - It's either Big-O or Big-Theta. If there is a case that it can grow slower than $G(n)$, use Big-O

Another Example

- Let's analyze `vector::push_back`

```
iterator insert(iterator it, const T& element) {
  size_t pos = it - begin();
  ensureCapacity(mSize + 1);
  for (size_t i = mSize; i > pos; i--) {
    mData[i] = mData[i-1];  // <-----  Most executed line: 1 (b)
  }
  mData[pos] = element;
  mSize++;
  return begin()+pos;
}
void push_back(const T& element) {
  insert(end(), element);
}

void expand(size_t capacity) {
  T *arr = new T[capacity]();
  for (size_t i = 0; i < mSize; i++) {
    arr[i] = mData[i];  // <-----  Most executed line: 0, 0, 0, 0, 2, 4, 8, 16 (a)
  }
  delete [] mData;
  mData = arr;
  mCap = capacity;
}

void ensureCapacity(size_t capacity) {
  if (capacity > mCap) {
    size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
    expand(s);
  }
}

// (a) can be 0 to n
```

```
// (b) can also be 0 to n
// Best case 0 + 0
// Worst case n + n
// vector::push_back is O(n)
```

Another Definition for $\mathcal{O}$ and Θ

- Using set builder notation
- $\mathcal{O}(g(n)) = \{f(n) \mid \text{there exists } c > 0 \text{ and } n_0 \geq 0 \text{ such that } f(n) \leq cg(n) \text{ for } n \geq n_0\}$
- $\Theta(g(n)) = \{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for } n \geq n_0\}$
- The result is the same as definition using lim

Summary

- We use Asymtotic Notation to describe efficiency of a program
  - Measure instruction count instead of time
  - Focus on growth rate of instruction count
- Find most frequently executed line in the code and count it
- Maps to Big-Theta if we have tight bound
- Use Big-O if we have upper bound

Example

Showing $\log_2(n!)$ is $\Theta(n \log_2(n))$

- Will use set definition of Big-Theta
  $\{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1 g(n) \leq f(n) \leq c_2 g(n) \text{ for } n \geq n_0\}$
- Need to find c_1, c_2 and n_0

Finding c_1 and c_2

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \cdots \times 1$$

$$n! \leq n \times n \times n \times n \times \cdots \times n$$

$\log n! \leq \log n^n$ $\quad \boxed{\log n^n = n \log n \text{ when } n \geq 1}$

Found $c_2 = 1$ for upper bound $\quad \boxed{\log n! \leq n \log n \text{ when } n \geq 1}$

$$n! = n \times (n-1) \times \cdots \times \left\lfloor \tfrac{n}{2} \right\rfloor \times \left(\left\lfloor \tfrac{n}{2} \right\rfloor - 1\right) \times \cdots \times 1$$

$$n! \geq \left\lfloor \tfrac{n}{2} \right\rfloor \times \left\lfloor \tfrac{n}{2} \right\rfloor \times \cdots \times \left\lfloor \tfrac{n}{2} \right\rfloor \times 1 \times \cdots \times 1$$

$n! \geq \left(\frac{n}{2}\right)^{n/2}$ $\quad \boxed{\log\left(\tfrac{n}{2}\right)^{n/2} = \tfrac{n}{2} \log n - \tfrac{n}{2}}$

$\log n! \geq \log\left(\frac{n}{2}\right)^{n/2}$ $\quad \boxed{0.5n \log n - 0.5n \geq 0.4n \log n \text{ when } n \geq 32}$

Found $c_1 = 0.4$ for lower bound $\quad \boxed{\log n! \geq 0.4n \log n \text{ when } n \geq 32}$

$\boxed{0.4n \log n \leq \log n! \leq n \log n \text{ when } n \geq 32, c_1 = 0.4, c_2 = 1, n_0 = 32}$

$\Longrightarrow \boxed{\log n! \text{ is } \Theta(n \log n) \text{ and } n \log n \text{ is } \Theta(\log n!)}$

Lecture 13

Priority Queue Simple Implementation

Featuring Binary Heap

Overview

- Simple implementation of `priority_queue`

- Quick intro to Graph and Tree
- Binary Heap
- priority_queue with Binary Heap

priority_queue

- Queue by value
- Max-in-First-Out

```
int main() {
  priority_queue<int> pq;
  pq.push(4);
  pq.push(20);
  pq.push(3);

  while (pq.empty() == false) {
    cout << pq.top() << endl;
    pq.pop();
  }
}
```

V0.1, priority_queue by vector

- Use vector to store data
- push = simply push_back
- top, pop = find the max value and return / erase
- max_element returns iterator to max element

```
namespace CP {
  template <typename T>
  class priority_queue {
    protected:
      std::vector<T> v;
    public:
      bool empty(); { return v.empty(); }
      bool size(); { return v.size(); }

      void push(const T &e) { v.push_back(e); }

      T top() { return *std::max_element(v.begin(), v.end()); }

      void pop() { v.erase(std::max_element(v.begin(), end())); }
  };
}
```

max_element

```
iterator max_element(iterator first, iterator last) {
  if (first == last) return last;
  iterator largest = first;
  ++first;
  for (; first != last; ++first)
    if (*largest < *first)
      largest = first;
  return largest;
}
```

V0.1 complexities

`push`	$\mathcal{O}(1)$ *amortized
`top()`	$\Theta(n)$
`pop()`	$\Theta(n)$